

PROGRAM 6: Deploy an automated build pipeline using docker hub

STEP 1: Create Project Directory

Create a directory and navigate into it:

```
mkdir prog6
```

```
cd prog6
```

STEP 2: Create Dockerfile

Example Dockerfile:

```
FROM node:18
```

```
WORKDIR /app
```

```
COPY . .
```

```
EXPOSE 3000
```

```
CMD ["node", "server.js"]
```

STEP 3: Build Docker Image

```
docker build -t prog6 .
```

STEP 4: Verify Image

```
docker images
```

STEP 5: Tag Image for Docker Hub

```
docker tag prog6 chandanas2173/prog6:latest
```

STEP 6: Login to Docker Hub

```
docker login -u chandanas2173
```

STEP 7: Push Image to Docker Hub

```
docker push chandanas2173/prog6:latest
```

```
chandanas2173/prog6:latest
1RV24MC027_CHANDANA_S@chandana-s-Inspiron-3593:~/program6$ docker push chandanas
2173/program6:latest
The push refers to repository [docker.io/chandanas2173/program6]
2a84bae126bc: Pushed
5fb15b7947e8: Layer already exists
7e0c7bda159a: Layer already exists
3b13f43fafc7: Layer already exists
d2a991bcab4d: Layer already exists
b624aa2d5ea2: Layer already exists
d399c9dc306f: Layer already exists
84f9fa179c1b: Layer already exists
ce84ba212e49: Layer already exists
e4dc8cd9ecc8: Layer already exists
6428cc293366: Layer already exists
2f7436e79a0b: Layer already exists
latest: digest: sha256:7594168138a36ce1856f12ea8693eba06d7a3d1d651ae193c486376dc
189c917 size: 2840
```

STEP 8: Generate GitHub Personal Access Token (PAT)

Go to GitHub → Settings → Developer Settings → Personal Access Tokens → Generate Token.

Enable the following scopes:

read:packages

write:packages

Copy your token.

STEP 9: Login to GitHub Container Registry (GHCR)

```
echo "<YOUR_GITHUB_PAT>" | docker login ghcr.io -u
iiAlmightii --password-stdin
```

STEP 10: Tag Image for GHCR

```
docker tag prog6 ghcr.io/chandanas2173/prog6:latest
```

STEP 11: Push Image to GHCR

```
docker push ghcr.io/chandanas2173/prog6:latest
```

STEP 12: Push Project Source Code to GitHub

```
git init
```

```
git add .
```

```
git commit -m "Program 6"
```

```
git branch -M main
```

```
git remote add origin
```

```
https://github.com/chandanas2173/program-6.git
```

```
git push -u origin main
```

OUTPUT:

Docker image successfully built, tagged, and pushed to Docker Hub and GHCR.

Project source code successfully uploaded to GitHub.

hub

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chandanas2173

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Repositories / prog6 / General

chandanas2173/prog6

Last pushed 27 days ago · 0 · 15

Add a description

Add a category

General

Tags

Image Management

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Webhooks

Settings

Tags

This repository contains 1 tag(s).

Tag	OS	Type	Pulled	Pushed
latest		Image	12 days	27 days

See all

Repository overview

INCOMPLETE

An overview describes what your image does and how to run it. It displays in the public view of your repository once you have pushed some content.

Add overview

Using 0 of 1 private repositories.

Docker commands

To push a new tag to this repository:

docker push chandanas2173/prog6:tagname

buildcloud

Build with Docker Build Cloud

Accelerate image build times with access to cloud-based builders and shared cache.

Docker Build Cloud executes builds on optimally-dimensioned cloud infrastructure with dedicated per-organization isolation.

Get faster builds through shared caching across your team, native multi-platform support, and encrypted data transfer - all without managing infrastructure.

Go to Docker Build Cloud

How likely are you to recommend Docker Hub to another developer?